

Trade Policy Analysis

Counterfactuals in general equilibrium: PTI, MTI, GETI

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Section 1

The counterfactual question

The world we do not observe

We observe the world **with** the agreements in place. The world **without** them exists in no dataset, so it has to be constructed from the model.

Two exercises follow from this, and they are the same computation run in opposite directions.

- **Ex post** — undo a policy that happened, and measure what it achieved.
- **Ex ante** — impose a policy that has not happened, and predict what it would do.

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The computation needs three ingredients.

- 1 The **trade-cost elasticity**, estimated in the gravity regression of the previous lecture. It tells us how much trade responds to a given change in costs.
- 2 The **structural equations**, which propagate that single change through every other country.
- 3 Trade and income data for one **reference year**, as a baseline.

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Notation

Write τ_{ij} for trade costs as they are, and τ_{ij}^c for the same costs in the **counterfactual** world without the agreement.

A **dot** denotes the proportional change between the two:

$$\dot{\tau}_{ij} \equiv \frac{\tau_{ij}^c}{\tau_{ij}}, \quad \dot{X}_{ij} \equiv \frac{X_{ij}^c}{X_{ij}}, \quad \dot{\Omega}_i \equiv \frac{\Omega_i^c}{\Omega_i}, \quad \dot{P}_j \equiv \frac{P_j^c}{P_j}$$

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Everything that follows is a statement about **ratios**, never about levels. We never need to know what trade would have been in dollars.

Three answers, not one

Applying the dot to the structural gravity equation:

$$\dot{X}_{ij} = \dot{\tau}_{ij}^{1-\sigma} \frac{\dot{Y}_i}{\dot{\Omega}_i^{1-\sigma}} \frac{\dot{E}_j}{\dot{P}_j^{1-\sigma}} \quad (1)$$

Each group of terms is a channel. Switching them on one at a time gives three different — and increasingly complete — answers.

GETI — incomes adjust as well

$\times \dot{Y}_i \dot{E}_j$

MTI — resistances adjust

$\times (\dot{\Omega}_i \dot{P}_j)^{\sigma-1}$

PTI — bilateral costs only

$\dot{\tau}_{ij}^{1-\sigma}$

Section 2

Layer 1 — the partial impact

Trade costs and the counterfactual

Suppose trade integration depends on distance and on the agreement:

$$\tau_{ij} = \kappa RTA_{ij} + \rho Dist_{ij} \quad (2)$$

From the gravity estimation we recover $\tau_{ij} = \exp[\hat{\kappa} RTA_{ij} + \hat{\rho} Dist_{ij}]$.

In the counterfactual world the agreement is gone, so $\tau_{ij}^c = \exp[\hat{\rho} Dist_{ij}]$.

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Dividing one by the other, everything except the policy cancels:

$$\dot{\tau}_{ij} = \exp[-\hat{\kappa} RTA_{ij}]$$

This is the **Partial Trade Impact**. Note what it costs us: we never had to know ρ , or the level of trade costs.

What the PTI misses

$$\dot{\tau}_{ij} = \exp[-\hat{\kappa} RTA_{ij}] \implies \dot{\tau}_{ij} = 1 \text{ for every non-member pair}$$

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The PTI says that a country which signed nothing experiences **exactly no change**. Its trade with every partner is unaffected.

We know from the model that this is false. Removing the agreements changes Ω_i and P_j **everywhere**, and therefore changes trade between countries whose own costs never moved.

The PTI is the answer the naive gravity equation gives. It is the first layer, and on its own it is the wrong answer.

Section 3

Layer 2 — the modular impact

What must move

To go further we need $\dot{\Omega}_i$ and $\dot{P}_j$: how the resistances themselves respond.

The recipe is mechanical.

- 1 With $\tau_{ij} = \exp[\hat{\kappa}RTA_{ij} + \hat{\rho}Dist_{ij}]$ and income data for a **reference year**, solve the system for Ω_i and P_j .
- 2 Replace τ_{ij} by τ_{ij}^c — the *only* thing that changes, incomes stay at their reference-year values — and solve **again** for Ω_i^c and P_j^c .
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So the counterfactual requires solving the price-index system twice. How is it solved at all?

Why there is no closed form

Each P_j is a **sum over every** Ω_i , each Ω_i a sum over every P_j , and the unknowns enter those sums **raised to the power** $1 - \sigma$.

With two countries a closed form does exist — you derived it in the exercise on multilateral resistances. With eighty-four countries there is no hope of writing one, and no need, since the system is solved numerically.

Two routes

Iterate. Guess $\Omega_i = P_j = 1$, compute one from the other, repeat until nothing moves. Head and Mayer (2014) embed this inside the estimation and iterate the whole thing — estimate, solve, re-estimate — which they call **SILS**, structurally iterated least squares.

Or avoid it. Fally (2015) gives conditions under which the resistances are *exactly* the fixed effects of a PPML regression. Anderson et al. (2018) build on this: estimate once by PPML, read the resistances off the fixed effects, and never solve a system. They call it **GEPPML**. See Yotov et al. (2016), and Herman (2021) for a Python implementation.

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Other routes exist: non-linear least squares under the price-index constraints, a log-linear Taylor expansion (Baier and Bergstrand 2009), or spatial-econometric approximations (Behrens et al. 2010).

A practical warning: normalisation

The system pins down the resistances only up to a scale, so one country must be **normalised**.

- In theory the choice is irrelevant. In practice all results are expressed relative to that country, so choose one with **reliable trade statistics**.
- Never normalise on a country your policy experiment actually hits — the counterfactual cannot be computed for the reference country.

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Rarely said out loud

These solvers often fail to converge. A small difference in the magnitude of a resistance can produce a large change in the function value. Rescaling the resistances usually fixes it. You will meet this in the code.

The modular trade impact

$$\text{MTI}_{ij} = \left[\frac{\dot{\tau}_{ij}}{\dot{\Omega}_i \dot{P}_j} \right]^{1-\sigma}$$

Incomes and expenditure are still held at their reference-year values — hence *modular*: the allocation of trade is solved in a different module from the determination of income (Anderson and Yotov 2010).

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Two things change relative to the PTI.

- For **members**, the effect shrinks. Removing the agreement makes them more remote from everyone, which raises their resistances and partly offsets the direct loss.
- For **non-members**, the effect is no longer one. They are affected although nothing changed in their own bilateral costs.

Section 4

Layer 3 — general equilibrium

Letting incomes move

Take labour as the only factor, $Y_i = w_i L_i$. With the labour force fixed, income changes are wage changes: $\dot{Y}_i = \dot{w}_i$.

Expenditure need not equal income — countries run deficits. Introduce them in an ad-hoc way, $E_j = (1 + d_j)Y_j$ with d_j constant (Dekle et al. 2008), so that

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We now need $\dot{Y}$. But trade depends on income, and income depends on trade. There is no way round solving them together.

The wage fixed point

Market clearing in changes gives, with $\lambda_{ji} = X_{ji}/Y_i$ the share of i 's expenditure spent on goods from j :

$$\dot{Y}_j = \left[\sum_i \frac{\lambda_{ji} (\dot{\tau}_{ji} \dot{Y}_j)^{1-\sigma}}{\sum_l \lambda_{li} (\dot{\tau}_{li} \dot{Y}_l)^{1-\sigma}} \dot{Y}_i Y_i \right] / Y_j \quad (3)$$

- $\dot{Y}$ appears on both sides. This is a **fixed point**, solved by the same kind of iteration as the resistances.
- The fraction is $\dot{\lambda}_{ji}$, the change in expenditure shares.

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Put $\dot{Y}$ and $\dot{E}$ back into Equation 1 and you have the **GETI** — the full general equilibrium answer.

Welfare

We could stop at trade flows. But the model gives welfare almost for free.

Using Shephard's lemma on P_j , the market clearing condition and zero internal trade costs, Costinot and Rodriguez-Clare (2014) show that the change in real consumption depends on **one number**:

$$\dot{C}_j = \lambda_{jj}^{\frac{1}{1-\sigma}}$$

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Why this is remarkable

Whatever happens to the thousands of bilateral flows, the welfare consequence for country j is summarised by how much more, or less, it buys from itself. This is also why the diagonal of the trade matrix is indispensable.

Section 5

Trade policy results

The experiment

Remove **every** regional trade agreement in the world, and compare.

- TradeProd data, year 2000, the 84 countries for which production — and therefore internal trade — is observed.
- Trade costs are not re-estimated: coefficients are the median structural estimates of the meta-analysis in Head and Mayer (2014), giving $\hat{\kappa} = 0.28$ for an RTA.
- Trade elasticity $\epsilon = -5.03$.

	PTI	MTI		GETI		Welfare	
Members:	Yes	Yes	No	Yes	No	Yes	No
RTA/FTA (all)	1.323	1.129	0.946	1.205	0.960	1.011	0.998

Median ratio of observed to counterfactual values. A value above one means the agreements raised the variable. Source: Head and Mayer (2014).

Reading the table

The multilateral adjustment is large. The partial impact says agreements raise members' trade by 32%. Letting the resistances respond cuts this to 13% — almost two thirds of the effect disappears once general equilibrium is respected.

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Income works the other way. The GETI, at 1.205, is *above* the MTI: members are richer with their agreements, and richer countries trade more.

Large trade effects, small welfare effects

Members gain **1.1%** of real income; non-members lose **0.2%**. An order of magnitude below the trade effects.

The reason is the sufficient statistic. Welfare moves only with $\dot{\lambda}_{jj}$, and most countries buy most of what they consume at home. A large proportional change in trade with one partner barely moves that share.

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Caveats

Much is missing from a model this simple. There is a single sector and a single σ for every good. There are no input-output linkages, so intermediates never cross a border twice. Firms are identical and none of them enters or exits. Labour is the only factor and it cannot move between countries. Deficits are held fixed by assumption.

Each of these matters for the size of the numbers, which is why Head and Mayer (2014) present them as an exposition of the method rather than as definitive estimates. See Costinot and Rodriguez-Clare (2014) for a fuller treatment of welfare.

Exercise

Open the folder **A4_GE**. In code there is a single R script, `pti_mti_geti.R`.

- 1 Read it beside the equations of this lecture and find the line that implements each: the contraction mapping for the resistances, the partial impact, and the wage fixed point of Equation 3.
- 2 Run it and check that you reproduce the table above.
- 3 Change the experiment — remove only the European Union, or double $\hat{\kappa}$. Which of the three measures moves most, and why?

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For research rather than teaching, the `gegravity` package of Herman (2021) implements this class of counterfactual in Python, and Yotov et al. (2016) provide a full course with data and programs.

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